

Table 1. HarmBench attack success rate by model scale, reported as full-sample means $\pm$ ten-group behavior-clustered jackknife standard errors. Human average is the equal-weight mean over the five frozen jailbreak templates; TBD denotes an incomplete model evaluation.

Method		Attack success rate (%) ↓						
		Direct	John	DNE	Jailbreak Bot	YOJA/Nona	APM	Human jailbreak average
Llama-3.2-1B-Instruct	Original	10.00 ± 2.08	5.83 ± 1.67	6.25 ± 1.55	4.58 ± 1.45	8.33 ± 1.08	2.92 ± 0.89	5.58 ± 0.77
	A-LQR	9.58 ± 1.76	6.25 ± 1.67	4.17 ± 1.08	3.33 ± 1.50	8.75 ± 1.45	5.00 ± 1.50	5.50 ± 0.25
	H _∞ (ours)	2.92 ± 1.40	1.25 ± 0.89	1.67 ± 0.68	0.42 ± 0.42	0.83 ± 0.56	0.42 ± 0.42	0.92 ± 0.26
Llama-3.2-3B-Instruct	Original	16.67 ± 2.48	11.67 ± 2.31	10.83 ± 2.08	14.17 ± 2.17	11.25 ± 1.53	10.83 ± 2.26	11.75 ± 1.07
	A-LQR	20.00 ± 2.90	11.67 ± 2.69	10.42 ± 2.17	12.92 ± 1.70	11.67 ± 1.04	11.25 ± 2.49	11.58 ± 1.32
	H _∞ (ours)	8.33 ± 1.86	4.17 ± 1.08	3.75 ± 0.97	3.75 ± 1.15	4.17 ± 0.88	2.50 ± 1.11	3.67 ± 0.43
Llama-3.1-8B-Instruct	Original	15.83 ± 4.02	23.33 ± 2.08	14.17 ± 4.31	9.17 ± 2.62	15.00 ± 4.78	11.67 ± 3.56	14.67 ± 2.03
	A-LQR	15.83 ± 3.82	15.00 ± 3.24	11.67 ± 4.68	7.50 ± 2.62	9.17 ± 3.15	10.83 ± 3.06	10.83 ± 1.99
	H _∞ (ours)	3.33 ± 1.84	1.67 ± 1.11	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.83 ± 0.83	0.50 ± 0.36